General Capabilities

Critical and creative thinking

Students develop critical and creative thinking as they learn to generate and evaluate ideas and possibilities when seeking new pathways or solutions. In the Science learning area, critical and creative thinking is embedded in the skills of questioning and predicting, solving problems through planning and conducting investigations, and analysing and evaluating evidence to make decisions and draw conclusions. Students develop an understanding of science concepts through inquiry that involves selecting appropriate information, evaluating sources of information to formulate hypotheses and reflecting on the processes used to reach evidence-based conclusions. Students' conceptual understanding becomes more sophisticated as they acquire an increasingly scientific view of the world and the ability to examine it from new perspectives.

Digital literacy

Students develop digital literacy as they operate and manage digital systems, practise digital safety and wellbeing, and while investigating, creating and communicating. In particular, they use digital literacy to access information; collect, analyse and represent data and information; model and interpret concepts and relationships; and communicate science ideas, processes and information.

Digital tools such as animations and simulation software can support student understanding of abstract phenomena, as they give opportunities to view phenomena and test predictions that cannot be investigated through practical investigations in the classroom.

Ethical understanding

Students develop their understanding of ethical concepts and ethical decision-making processes in relation to science investigations, codes of practice, and the use of scientific information and science applications. They learn about ethical procedures for investigating and working with people, animals, data and materials. Students use scientific information to evaluate claims and to inform ethical decisions about a variety of social, environmental and personal issues. They consider their own roles as discerning citizens and learn to analyse biases and assumptions as they apply ethical concepts when making decisions in complex situations.

Literacy

In Science, students develop literacy capability as they explore and investigate. They comprehend and compose texts, including those that give information; describe events and phenomena; recount experiments; present and evaluate data; give explanations; and present ideas, opinions and claims. They comprehend and compose multimodal texts, such as charts, graphs, diagrams, pictures, maps, animations, models and visual media. Language structures and text structures are used to link information and ideas, give descriptions and explanations, formulate hypotheses and construct evidence-based arguments capable of expressing an informed position.

Scientific vocabulary is often technical and includes specific terms for concepts and features of the world, as well as terms that encapsulate an entire process in a single word, such as photosynthesis. Language is therefore essential in providing the link between the concept itself and student understanding, and assessing whether the student has understood the concept.

Numeracy

Students develop and use numeracy through investigation of science concepts and application of science inquiry practices. The Science understanding contentis closely linked to numeracy through the focus on scale and measurement, and patterns, order and organisation.

Students develop numeracy by focusing on measurement and data collection when applying science inquiry practices. They identify patterns in data and use mathematical relationships to represent those patterns. They represent primary and secondary data using tables, displays and visualisations, and interpret data to construct evidence-based arguments and conclusions. In later years, they engage in statistical analysis of data and consider issues of validity and reliability of data.

Personal and social capability

Students develop self-awareness and self-management skills as they direct their own learning, plan and carry out investigations, and become independent learners who can apply science understanding and practices to make decisions. They build skills in social awareness and social management as they engage in collaborative investigations that require them to work cooperatively in teams, share resources and processes, make group decisions and show leadership. Empathy and respect are developed as students identify and learn about diverse world views and perspectives that have informed the development of science, and ways in which different individuals and groups may perceive scientific knowledge, advances or solutions.

Pre-primary

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning builds on the Early Years Learning Framework and each child's funds of knowledge. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Science provides opportunities for children to explore, make observations and develop their understanding of their world.

In Pre-primary, children recognise that plants and animals share the same basic needs, and these can be met by the places where they live. They observe everyday objects and the materials from which they are made. Children explore daily and seasonal changes and changes in the world around them. They explore factors that affect how objects move.

Children learn that observations and experiences can be used to pose questions and make predictions. They participate in investigations, share their questions, predictions and ideas with others and develop scientific ideas about the natural and physical world.

Achievement Standard

By the end of the year:

Children describe the basic needs of plants and animals and how these are met by the places they live. They describe the observable properties of various materials used in everyday objects. Children describe how daily and seasonal changes in the environment affect everyday life. They describe the influence of size, shape, material and force on the movement of objects.

Children in Pre-primary pose questions and make predictions based on their shared experiences and knowledge. They use their senses in guided investigations to make observations, discuss these and represent them in a variety of ways. Children discuss similarities between their predictions and observations.

Content Descriptions

Science understanding

Biological sciences

Plants and animals have basic needs that are met by the places they live

For example:

• basic needs include air, food, water and shelter

• the places plants and animals live include our homes and local areas, wetlands, forests, deserts and oceans

• how connection to Country, roles and responsibilities to care for plants and animals are important to Aboriginal and Torres Strait Islander peoples

WAPSSUB1

Critical and creative thinking

Chemical sciences

Objects are made of various materials that have observable properties

For example:

• observable properties of materials include colour, hardness, texture and flexibility

WAPSSUC1

Numeracy Critical and creative thinking

Earth and space sciences

Daily and seasonal changes in the environment affect our local community and the world around us

For example:

• daily changes, such as temperature, wind strength and direction, rain and cloud cover, and how humans, plants and animals respond to them

• seasonal changes, such as flowering and fruiting, activity of insects and bird behaviour

• the seasonal calendars of Aboriginal and Torres Strait Islander peoples are based on visible changes in the natural environment

WAPSSUE1

Numeracy

Physical sciences

The way objects move depends on factors, including their size, shape, material and the force applied

For example:

• comparing the movement of similar-shaped, but different sized objects, such as marbles and basketballs

• the movement of traditional Aboriginal and Torres Strait Islander peoples' tools, such as boomerangs and digging sticks

WAPSSUP1

Numeracy Critical and creative thinking

Science inquiry

Questioning and predicting

Pose questions and make predictions based on prior knowledge and shared experiences

For example:

- posing questions, such as
  - what shape of block will roll down the ramp?
  - will it rain today?
- making predictions, such as
  - when we visit the local wetland, I predict we will see plants and animals

WAPSSIQ1

Numeracy Critical and creative thinking

Planning and conducting

Participate in guided and self-initiated investigations safely

For example:

- testing ideas using the five senses, such as how different surfaces are cool or warm to touch
- discussing ways to conduct investigations safely

WAPSSIPL1

Numeracy Critical and creative thinking

Make observations using comparison

For example:

- using pictures and drawings to represent observations
- discussing observations with others in guided conversations

WAPSSIPL2

Critical and creative thinking

Processing, modelling and analysing

Represent and discuss observations and identify patterns

For example:

- discussing places where animals live and identifying patterns in their needs and features of their habitat
- counting the number of drink bottles in the class made from different materials and representing this information using a picture graph

WAPSSIPR1

Numeracy Critical and creative thinking

Evaluating

Discuss similarities and differences between predictions and observations

For example:

- discussing and comparing their observations of the weather in the afternoon to predictions made in the morning
- predicting the plants and animals that will be found on a field trip to a local habitat and comparing these to observations

WAPSSIE1

Critical and creative thinking

Communicating

Share questions, predictions, observations and ideas with others

For example:

- describing observations to others through discussions, drawings or models
- retelling the steps of an investigation and what happened

WAPSSICM1

Literacy Personal and social capability

Collaborating and applying

Use the senses to learn about the natural and physical world and develop scientific ideas

For example:

• responding to questions posed by teachers or children about the way different materials behave
• interacting with stories or documentaries about scientists, such as Dame Jane Goodall or Sir Joseph Banks and noticing the ways they make their observations, such as through drawings, collections, sound recordings and photography, and how they ask questions about what they think they will observe and find

WAPSSICL1

Critical and creative thinking

Year 1

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Science provides opportunities for children to use their senses to observe and gather information, describe, make comparisons, sort and classify to create an order that is meaningful.

In Year 1, children group plants and animals based on their observations of external features. They observe physical changes to materials to establish their composition remains unaffected. Children explore water as a natural resource and how it is used by people, plants and animals in different ways. They investigate how objects can move or change shape when force is applied.

Children pose questions and make predictions and are introduced to ways of sorting and ordering data. They compare their observations with their predictions and communicate findings with others. Children use their science knowledge to make decisions and choices in their environment, such as how to stop an object in motion or how to look after water sources.

Achievement Standard

By the end of the year:

Children group plants and animals by observing their external features. They understand that physical changes to materials do not affect their composition. Children recognise that water is a natural resource that comes from a variety of sources and is used in different ways by plants, animals and people. They describe the strength and direction of a push or pull force and how it affects the motion and/or shape of an object.

Children pose questions to explore science ideas and make predictions based on experiences and prior knowledge. They engage safely in guided investigations, making and recording observations, including informal measurements. Children use provided tables to sort data and represent it visually and with physical models. With guidance, children compare observations with predictions and identify further questions. They use everyday vocabulary to communicate observations and findings. Children use their science knowledge to help them make choices and decisions, such as choosing ways to manipulate materials to alter their appearance.

Content Descriptions

Science understanding

Biological sciences

Plants and animals have external features that serve a purpose and by which they can be grouped

For example:
• the external features of plants can include leaves, stems, flowers, fruits and roots
• the external features of animals can include eyes, body coverings, legs, wings, fins, antennae and claws
• the body coverings of various groups of animals, such as fur, scales and feathers
• Aboriginal and Torres Strait Islander peoples' use of tracks to identify, locate and follow animals

WA1SSUB1

Numeracy Critical and creative thinking

Chemical sciences

Materials can be changed physically without changing their composition

For example:
• materials can be physically changed to suit a particular purpose, such as twisting strands of cotton or wool together to make thread stronger
• physical changes can include scrunching, twisting, pulling, stretching, melting, bending or breaking into smaller pieces by cutting, tearing or crushing

WA1SSUC1

Critical and creative thinking

Earth and space sciences

Water is a natural resource that comes from a range of sources and is used by people, plants and animals in different ways

For example:

- the key processes of the water cycle are evaporation (drying), condensation (clouds) and precipitation (rain)
- plants and animals require water as a habitat and to maintain life
- Aboriginal and Torres Strait Islander peoples' connections with and valuing of water and water resource management

WA1SSUE1

Critical and creative thinking

Physical sciences

The strength and direction of a push or a pull force affects how an object moves or changes shape

For example:

- pushing or pulling on an object can start or stop its motion, change its direction and speed of travel, or change its shape
- push and pull forces can be represented using models, drawings or role-play
- traditional games, such as kendama, Daruma Otoshi, marbles and four-square handball use push and pull forces

WA1SSUP1

Numeracy Critical and creative thinking

Science inquiry

Questioning and predicting

Pose questions and make predictions based on knowledge and experiences

For example:

- posing questions, such as
  - what will happen if I pull playdough hard?
- making predictions, such as
  - if we leave a plate of water on a shelf, the water will disappear

WA1SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Engage in guided investigations to answer questions, test predictions, and assess risks

For example:

- suggesting ways to conduct investigations safely, including wearing safety glasses and aprons
- comparing how different objects move when pushed or pulled

WA1SSIPL1

Literacy Critical and creative thinking

Make and record observations, including informal measurements

For example:

- making observations using the senses to gather information and record it using text, drawing, counts or digital tools
- using uniform informal units of measurement, such as handspans, walking paces, blocks or popstick lengths

WA1SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Sort and order data using provided tables and represent data using visual or physical models

For example:

- representing the number of plants or animals in a collection using a picture graph
- using digital images to show how pushes and pulls affect the shape of an object and sorting images into before and after columns of a table

WA1SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare observations to predictions and identify further questions for investigation

For example:

• comparing observations of movement with predictions, such as how far an object may travel

• comparing findings of water use surveys and discussing ways to investigate the differences

WA1SSIE1

Literacy Critical and creative thinking

Communicating

Communicate observations, ideas, and findings using everyday and scientific vocabulary

For example:

• drawing or modelling the external features of a plant or animal and labelling with scientific terms

• representing push and pull forces using role-play, labels, arrows or time lapse drawings and describing their representation

WA1SSICM1

Literacy Personal and social capability

Collaborating and applying

Use science knowledge and understandings to make decisions and choices in their environment

For example:

• describing the best way to push and pull to complete a ball and cup game

• recommending strategies to reduce water usage at school

WA1SSICL1

Numeracy Critical and creative thinking

Year 2

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Science provides opportunities for children to use their senses to observe and gather information, describe, make comparisons, sort and classify to create an order that is meaningful.

In Year 2, children explore the life cycles of plants and animals and the ways in which they grow, change and produce offspring. They build on their understanding of the properties of materials to recognise the purposes they serve and how the properties change when materials are combined. Children learn that Earth is a planet in the solar system that orbits a star (the Sun). They recognise that sound energy is produced by a variety of sources.

Children build on their experiences of the physical world by engaging in guided investigations to explore ideas and answer questions. They make and record observations using informal measurements, and sort and order data using provided tables. They compare their observations with their predictions and identify further questions. Children use scientific vocabulary to explain observed phenomena and make decisions in their environment.

Achievement Standard

By the end of the year:

Children identify and sequence the life stages of specific plants and animals. They describe ways materials can be combined for different purposes. Children represent Earth's position in the solar system and its orbit around a star (the Sun). They recognise that sound is a form of energy produced by natural and human sources and can be sensed.

Children pose questions to explore science ideas and make predictions based on experiences and prior knowledge. They make and record observations, including informal measurements, during guided investigations. They use provided organisers to sort, order and represent patterns in data. Children compare their observations with those of others and identify further questions. They use everyday and scientific vocabulary to communicate observations, findings and ideas. Children use their science knowledge to help them make choices and decisions, such as selecting materials based on their ability to produce certain sounds.

Content Descriptions

Science understanding

Biological sciences

Plants and animals have life cycles through which they grow, change and have offspring

For example:

• living things have predictable characteristics at different stages of development, such as egg, larva, pupa, adult in insects

• flowering plants have predictable life cycles and reproduce using seeds

WA2SSUB1

Critical and creative thinking

Chemical sciences

Materials can be combined for a particular purpose

For example:

• combining materials can change their properties, such as mixing water and cornflour to make a non-Newtonian fluid

• the ways in which Aboriginal and Torres Strait Islander peoples combine different materials to produce tools, including hafting, weaving and gluing

WA2SSUC1

Critical and creative thinking

Earth and space sciences

Earth is a planet in the solar system that orbits a star (the Sun)

For example:

• the revolution of Earth around the Sun

• the solar system contains celestial objects, including planets, moons , asteroids and the Sun

• patterns in the sky and their use by Aboriginal and Torres Strait Islander peoples

WA2SSUE1

Numeracy Critical and creative thinking

Physical sciences

Sound energy is produced by a range of natural and human-made sources and can be sensed

For example:

• ways to produce sound using familiar objects and actions, such as striking, blowing, scraping, plucking and shaking, and the qualities of sounds, including volume and pitch

• sound is transmitted, absorbed and reflected differently by various materials and environments, affecting how it is heard and perceived

• traditional Aboriginal and Torres Strait Islander peoples' instruments, how they make their characteristic sounds and their use in communications and ceremonies

WA2SSUP1

Literacy Critical and creative thinking

Science inquiry

Questioning and predicting

Pose questions and make predictions based on knowledge and experiences

For example:

• posing questions, such as
  • how long will it take for our silkworm eggs to hatch?

• making predictions, such as
  • if I loosen a string it will make a sound with a lower pitch

WA2SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Engage in guided investigations to answer questions, test predictions, and assess risks

For example:

• demonstrating appropriate use of materials and equipment

• following visual or verbal steps to construct a musical instrument

WA2SSIPL1

Literacy Critical and creative thinking

Make and record observations, including informal measurements

For example:

• making observations of the different life stages of silkworms or a germinating seed and recording the change of appearance by taking digital images and providing a description
• observing and drawing the day and night sky and comparing the two

WA2SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Sort and order data using provided tables and represent data using visual or physical models

For example:
• using a graphic organiser to sort images of musical instruments and the actions used to produce their sound
• making and labelling models of the observed stages in the life cycle of a silkworm or a germinating seed

WA2SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare observations to predictions and identify further questions for investigation

For example:
• comparing the result of combining two materials, such as cornflour and water, to predictions
• comparing sounds made by percussion instruments to predictions

WA2SSIE1

Literacy Critical and creative thinking

Communicating

Communicate observations, ideas, and findings using everyday and scientific vocabulary

For example:
• sequencing the stages in the life cycle of a plant or animal and using scientific terms to label them
• representing the steps in an investigation using a series of captioned digital images

WA2SSICM1

Literacy Personal and social capability Digital literacy

Collaborating and applying

Use science knowledge and understandings to make decisions and choices in their environment

For example:
• constructing an instrument made from materials and making choices based on the sounds and properties of each
• investigating the stories of people who used multiple observations to develop scientific explanations, such as 17th century entomologist and naturalist Maria Sibylla Merian, who was the first to record the nature of metamorphosis

WA2SSICL1

Critical and creative thinking

Year 3

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Science provides opportunities for students to explore their physical, social, cultural and technological surroundings. As students develop the ability to work collaboratively, they work with others to plan and make decisions in constructing knowledge.

In Year 3, students observe the characteristics of living things and use these to group them. They apply criteria to identify things as living, non-living or once-living. They recall the observable properties of solids and liquids and understand that a change of state is caused by adding or removing heat. Students identify important Earth resources and how humans and other living things use them in interconnected ways. They learn energy can be transferred and transformed.

Students pose questions and make predictions. They plan and conduct fair investigations and compare their findings with others, and to their predictions. They represent observations and data using simple tables and column graphs or other visual or physical models and communicate ideas and findings using scientific vocabulary. Students use science knowledge to propose explanations for observed phenomena and solutions to problems.

Achievement Standard

By the end of the year:

Students group living things based on their characteristics and distinguish them from non-living and once-living things. They recall the observable properties of solids and liquids and understand that a change of state in a substance results from adding or removing heat. Students recognise that soils, rocks and minerals are important Earth resources and are used by both humans and other living things in different and interconnected ways. They understand that energy can be transferred and transformed.

Students explore science ideas by posing guided questions about the effect on one variable of changing another. They make predictions, with assistance, and test them by planning and conducting safe investigations that account for variables to be kept the same. They make and record observations with some familiar scaled instruments, such as thermometers. They organise data using tables, column graphs or models. With support, they communicate findings using familiar scientific vocabulary. Students use science knowledge to propose explanations for observed phenomena and simple solutions to problems.

Content Descriptions

Science understanding

Biological sciences

Living things can be distinguished from non-living and once-living things, and grouped by their characteristics

For example:

• the characteristics of vertebrates, including fish, amphibians, reptiles, birds and mammals, and invertebrates, including insects, molluscs and arachnids, can be used to classify them using a dichotomous key

• the criteria for something to be recognised as living may be recalled with the mnemonic MRS GREN

• systems of grouping living things used by Aboriginal and Torres Strait Islander peoples, such as by their purpose, age, stage in life cycle

WA3SSUB1

Critical and creative thinking

Chemical sciences

The observable properties of solids and liquids and how adding or removing heat leads to a change of state

For example:

• freezing and solidification are caused by removing heat

• melting is caused by adding heat

• a solid will hold its shape while a liquid will fill the bottom of a container

WA3SSUC1

Critical and creative thinking

Earth and space sciences

Soils, rocks and minerals are important Earth resources, and are used by humans and other living things in different and interconnected ways

For example:

• the use of soil by humans and other living things, such as for food, nutrients and habitat

• minerals extraction methods, the potential environmental and ecological impacts, and the strategies used to mitigate these

• the properties of local soil types, including colour, depth, texture and structure

• Aboriginal and Torres Strait Islander peoples' connection to and respect for land and their use of its resources, such as using ochre for painting and rock for grinding and tools

WA3SSUE1

Critical and creative thinking

Physical sciences

Energy can move from one thing to another (transfer), and change form (transform)

For example:

• energy transformation, such as from electrical energy to heat in a heater, or chemical energy to movement energy when running

• energy transfer, such as using a heat source to transfer heat to an ice cube, or kicking a ball

WA3SSUP1

Critical and creative thinking

Science inquiry

Questioning and predicting

Pose questions and make predictions based on planned observations of phenomena that include variables to be measured and changed

For example:

- posing questions, such as
  - how fast will ice melt if we place it in the sun?
  - how high will a plant grow in a week if we plant it in potting mix?
- making predictions, such as
  - seeds planted in potting mix will germinate faster than in sandy soil

WA3SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct investigations, including elements of fair tests, and consider the material and equipment risks

For example:

- using a planner in a teacher-led activity to identify what to change, what to keep the same and what to measure in an investigation

WA3SSIPL1

Literacy Critical and creative thinking

Make and record observations, including formal measurements using familiar scaled instruments

For example:

- exploring how to make readings with equipment, such as thermometers or measuring cylinders
- recording measurements made with familiar scaled instruments using tables or graphic organisers

WA3SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Organise and represent data using tables, column graphs and models to identify patterns

For example:

- creating a visual representation of the layers of soil observed in the school garden

WA3SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare findings with those of others, and to predictions; consider if investigations were fair; and identify questions for further investigation

For example:

- discussing the factors that make investigations fair and evaluating the fairness of their own and others' investigations

WA3SSIE1

Literacy Critical and creative thinking

Communicating

Communicate ideas using scientific vocabulary

For example:

- discussing how to prepare simple reports of their investigations to share predictions, methods, results and conclusions with their peers
- representing energy transfer using diagrams, digital drawings, arrows or labels using scientific vocabulary

WA3SSICM1

Literacy Personal and social capability Digital literacy

Collaborating and applying

Use science knowledge to propose explanations for observed phenomena and solutions to problems

For example:

- observing evaporation and investigating ways to keep soil moist while conserving water
- analysing how data collected on the transfer of heat can be used to determine the best conductors or insulators

WA3SSICL1

Critical and creative thinking

Year 4

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Science provides opportunities for students to explore their physical, social, cultural and technological surroundings. As students develop the ability to work collaboratively, they work with others to plan and make decisions in constructing knowledge.

In Year 4, students represent the relationships between consumers, producers and decomposers using food chains. They investigate different materials and their properties and relate these to their use. Students appreciate that Earth's surface changes over time due to a variety of processes. They observe and investigate forces that operate from a distance, such as magnetism, and direct contact, such as friction.

Students pose questions involving a changed variable and make predictions using their observations and science knowledge. Students plan investigations that include elements of fair tests and consider the material and equipment risks. They make and record observations and use formal measurements and familiar scaled instruments to collect and record data that they organise and represent using tables and column graphs to identify patterns. Students use science knowledge to propose explanations and solutions to problems and identify questions for further investigation.

Achievement Standard

By the end of the year:

Students describe the interaction between producers, consumers and decomposers in an ecosystem and represent these using food chains. They connect raw materials to the processed materials that are made from them and relate the use of materials to their observable properties. Students identify the processes that result in slow and rapid changes to Earth's surface over time. They identify contact and non-contact forces and describe, with examples, how they affect the behaviour of objects.

Students explore science ideas by posing questions, with guidance, that include variables to measure and change. They make predictions based on observations and test them by planning and conducting scaffolded investigations. They make and record observations in a variety of ways and using a diversity of scaled instruments. They organise and represent data in tables, column graphs and models to identify patterns. They communicate findings using scientific vocabulary in correct contexts. Students use science knowledge to propose explanations for observed phenomena and solutions to problems.

Content Descriptions

Science understanding

Biological sciences

Producers, consumers and decomposers have roles within an ecosystem and interact in ways that can be represented by food chains

For example:

• the Sun's role as the primary energy source in food chains, provided through photosynthesis in producers

• representing the transfer and direction of the flow of energy within an ecosystem with arrows

• exploring local ecosystems, recognising producers, consumers and decomposers and representing these using a food chain

• Aboriginal and Torres Strait Islander peoples' knowledge and use of food chains

WA4SSUB1

Critical and creative thinking

Chemical sciences

Processed materials, including fibres, metals, glass and plastics, are made from raw materials, such as wool, ores, sand and oil, and have a range of physical properties that influence their use

For example:

• the physical properties of materials include flexibility, hardness, strength, absorbency and weight

• reasons why materials, or combinations of materials, are used in familiar objects, such as shoes, drink containers or backpacks

• the use of raw and processed materials by Aboriginal and Torres Strait Islander peoples for different purposes, such as tools, clothing and shelter

WA4SSUC1

Critical and creative thinking

Earth and space sciences

Weathering, erosion, transportation and deposition cause slow or rapid change to Earth’s surface

For example:

• weathering caused by mechanical means, such as wind abrasion and freeze-and-thaw; chemical means, such as acid rain; and biological means, such as by plants and tree roots

• landforms created by these processes, such as dunes, canyons and caves

• the significance to Aboriginal and Torres Strait Islander peoples of landforms, such as Katter Kich (Wave Rock) and its importance to Ballardong Noongar people

WA4SSUE1

Critical and creative thinking

Physical sciences

Forces are exerted by one object on another through direct contact, such as friction, or from a distance, such as magnetism and gravity

For example:

• gravity is a non-contact force that pulls all objects with mass towards each other, regardless of whether they are moving or not moving

• magnetic force is a non-contact force and can attract (pull) and repel (push) objects at a distance

• friction is a contact force that occurs when two things touch and move against each other and slow down or stop moving

• force arrows represent the direction and strength of forces acting on an object

WA4SSUP1

Critical and creative thinking

Science inquiry

Questioning and predicting

Pose questions and make predictions based on planned observations of phenomena that include variables to be measured and changed

For example:

• posing questions, such as
  • what will happen to the distance a toy car travels if the wheels are changed?
  • which material will make the best drink bottle?

• making predictions, such as
  • the number of producers will increase if herbivores are removed
  • as the slope gets steeper more erosion will occur

WA4SSIQ1

Critical and creative thinking

Planning and conducting

Plan and conduct investigations, including elements of fair tests, and consider the material and equipment risks

For example:

• planning a fair test, using a planner to test the amount of grip (friction) of a range of shoes

WA4SSIPL1

Critical and creative thinking

Make and record observations, including formal measurements using familiar scaled instruments

For example:

• making measurements with a variety of scaled instruments, such as force meters

• recording observations in various ways, including photographs, videos, labelled drawings and tables

WA4SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Organise and represent data using tables, column graphs and models to identify patterns

For example:

• modelling the number of producers and consumers in an ecosystem and how these change over time

• graphing the volume of liquid absorbed by a variety of materials

WA4SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare findings with those of others, and to predictions; consider if investigations were fair; and identify questions for further investigation

For example:

• comparing own findings from investigations with others and asking questions about factors that may have led to any differences in findings

• identifying unexpected findings and posing questions for further investigation

WA4SSIE1

Literacy Critical and creative thinking

Communicating

Communicate ideas using scientific vocabulary

For example:

• modelling landscapes using materials, such as sand, gravel, soil and rocks to show labelled effects of erosion by water

• representing the direction and strength of forces using arrows, and labelling the forces

WA4SSICM1

Literacy Personal and social capability Digital literacy

Collaborating and applying

Use science knowledge to propose explanations for observed phenomena and solutions to problems

For example:

• designing an investigation to test how storage life for fruit may be increased

• examining the wheels of different toys and evaluating which material has the most friction on different surfaces

WA4SSICL1

Critical and creative thinking

Year 5

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Science provides opportunities for students to explore their physical, social, cultural and technological surroundings. As students develop the ability to work collaboratively, they work with others to plan and make decisions in constructing knowledge.

In Year 5, students describe the structural and behavioural adaptations of living things that enable them to survive in their environment. They model the motion and arrangement of atoms and molecules (particles) to explain observable properties of matter. Students model the relationship between the Sun and the planets in the solar system and illustrate how Earth's rotation on its axis and revolution around the Sun relate to observable cyclic phenomena. They explore light and recognise that it can be absorbed, reflected, refracted and form shadows.

Students pose testable questions that include variables to be changed and measured. They plan and conduct safe investigations and use equipment to measure and record data. Students organise and represent data to identify the relationships between variables that are measured and changed. They compare their methods and findings with those of others, and to their predictions, and report on their investigations using suitable language features. Students use science knowledge to develop considered responses to problems, at a local and global level, through investigation and research.

Achievement Standard

By the end of the year:

Students describe the structural and behavioural adaptations of various living things that enable survival in their environment. They describe and compare the observable properties and behaviours of solids, liquids and gases, and relate this to the arrangement of atoms and molecules (particles). Students model the relationship between the Sun and the planets in the solar system and illustrate how Earth's rotation on its axis and revolution around the Sun relate to observable cyclic phenomena. They identify sources of light, describe how light travels in a straight line and explain how it is reflected, refracted, absorbed, forms shadows and can be sensed.

Students pose testable questions that include variables to be changed and measured. They apply science knowledge to make predictions and, with guidance, plan and conduct safe, fair investigations to test them. Students use a variety of equipment to observe, measure and record data that they organise using tables, graphs and models to identify the relationship between variables, including using line graphs to represent continuous data. Students compare their methodology with others to determine whether investigations were fair and identify ways to improve their process. They communicate their investigation procedure, data and results in various ways. Students use science knowledge to develop considered responses to problems, at

a local level, through investigation and research.

Content Descriptions

Science understanding

Biological sciences

Living things have structural and behavioural adaptations that enable their survival in their environment

For example:

• the adaptations of animals and plants to Australia's desert environments, such as the large ears and nocturnal habit of the bilby, or thin, spiky leaves and deep root systems of desert plants, such as spinifex

• the adaptations of animals and plants in ocean environments, such as humpback whales' communication patterns to navigate and hunt prey, or the horizontal rhizomes and air canal containing leaves of seagrass

• camouflage is an adaptation to hide from predators and/or ambush prey

WA5SSUB1

Critical and creative thinking

Chemical sciences

The observable properties of solids, liquids and gases can be explained by the motion and arrangement of atoms and molecules (particles)

For example:

• substances can be classified as solids, liquids or gases based on their properties, including their ability to hold or change shape or volume and their ability to flow

• the arrangement of atoms and molecules in each state of matter can be represented using drawings or models

• a change of state between solid, liquid and gas is caused by adding or removing heat; this heat change affects how fast the particles move and the distance between them

WA5SSUC1

Critical and creative thinking

Earth and space sciences

The movement of Earth and other planets relative to the Sun and how Earth's rotation on its axis and revolution around the Sun relate to cyclic observable phenomena, including the day/night cycle

For example:

• the revolution and rotation of the planets of the solar system

• the apparent movement of celestial objects, such as the Moon and constellations, through the sky

• seasons are due to the tilt of Earth's axis altering the hours of direct sunlight on Earth

• Aboriginal and Torres Strait Islander peoples' understanding of the night sky and its use for timekeeping, navigation and storytelling

WA5SSUE1

Critical and creative thinking

Physical sciences

Light energy travels from a source in a straight path and can be absorbed, reflected, refracted, form shadows and be sensed

For example:

• materials selectively absorb, reflect or transmit light of different colours

• opaque, translucent and transparent materials allow the transmission of light to differing degrees

• light enters the eye and is refracted by the lens onto the retina, where it is sensed

WA5SSUP1

Critical and creative thinking

Science inquiry

Questioning and predicting

Pose testable questions that include variables to be measured and changed, and apply science knowledge to make predictions

For example:

• posing questions, such as

  • how will the diameter of a balloon filled with air be affected when the temperature of the area around it changes?

  • what is the difference between the length of day and night at different times of the year?

• making predictions, such as

• shadows cast by a piece of dowel inserted vertically into the ground will grow shorter towards the middle of the day

WA5SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct fair, safe and repeatable investigations

For example:

• with guidance, planning and recording the method to be used in an investigation so that it can be repeated

• making decisions on the variables to be controlled in fair tests

WA5SSIPL1

Literacy Critical and creative thinking Ethical understanding

Use equipment to observe, measure and record data

For example:

• comparing the accuracy of different measuring instruments, such as a measuring jug and a cup, and discussing why accuracy is important

• recording data using standard units, such as grams, seconds and metres

WA5SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Organise and represent data using tables, graphs and models to identify the relationships between measured and changed variables

For example:

• representing diagrammatically the connection between animals, their environments and adaptations

• modelling the movement of the planets in the solar system

WA5SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare findings with those of others, and to predictions; evaluate the fairness of an investigation and suggest improvements; and pose questions for further investigation

For example:

• comparing methods and findings with those of others and/or to their own predictions to determine if the investigation was a fair test

• identifying the strengths and weaknesses of their own and others' investigations, and improvements that can be made

WA5SSIE1

Literacy Critical and creative thinking Personal and social capability

Communicating

Communicate ideas in a variety of ways, including scientific reports with appropriate language features

For example:

• preparing a scientific report for an investigation using scientific vocabulary and data representations, such as tables and graphs

• annotating digital photography or field sketches to describe structural adaptations of plants or animals

WA5SSICM1

Literacy Personal and social capability Digital literacy

Collaborating and applying

Use science knowledge to develop considered responses to problems, at a local and global level, through investigation and research

For example:

• applying knowledge of animal adaptations to propose solutions to prevent the extinction of an endangered species

• participating in a citizen science project in the local community

WA5SSICL1

Critical and creative thinking Personal and social capability

Year 6

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Science provides opportunities for students to explore their physical, social, cultural and technological surroundings. As students develop the ability to work collaboratively, they work with others to plan and make decisions in constructing knowledge.

In Year 6, students explore the relationship between the growth and survival of living things, changing conditions in their environment and the influence of human activities. They are introduced to ways to classify changes to substances as reversible and irreversible. Students explore the effects of sudden geological changes and extreme weather on the surface of Earth. They identify the role of the components in electrical circuits and learn to describe energy flows in terms of transfer and transformation.

Students pose questions that include the variable to be changed and apply science knowledge to make reasoned predictions of the results. To test their predictions, they plan and conduct fair, repeatable investigations that include controlled variables. Students describe the observed relationships in the collected data between variables to be measured and changed, compare their results to their predictions and pose further questions to investigate. Students report on their investigation and findings using appropriate language features. They use science knowledge to develop considered responses to problems, at a local and global level, through investigation and research.

Achievement Standard

By the end of the year:

Students explain how changes in conditions in the environment and the influence of human activities affect the growth and survival of living things. They classify and compare reversible and irreversible changes to substances. Students describe the effect of sudden geological changes and extreme weather on Earth's surface. They identify the role of circuit components, insulators and conductors in the transfer and transformation of electrical energy.

Students pose testable questions that include variables to be changed and measured and apply science knowledge to make reasoned predictions. They plan and conduct safe, fair, repeatable investigations to test predictions and identify the relationship between variables. Students use equipment to generate and record data that they process and represent using tables, graphs and models to describe their findings, including using line graphs to represent continuous data. They evaluate the fairness of an investigation, identifying difficulties and ways to minimise these. They compare their results to their predictions, posing questions for further investigation. With assistance, students draw conclusions based on collected data and communicate these in various ways, including scientific reports. They use science knowledge to develop considered responses to problems, at a local and global level, through investigation and research.

Content Descriptions

Science understanding

Biological sciences

The growth and survival of living things are affected by the changing conditions in their environment and the influence of human activities

For example:

• the cause and effect of changes to salinity, substrate, sunlight or temperature, on plants and microorganisms

• the effect of increased sea temperature on coral reefs, such as bleaching

• Aboriginal and Torres Strait Islander peoples' observations of plant and animal responses to seasonal change allows them to effectively manage and farm food resources

WA6SSUB1

Critical and creative thinking

Chemical sciences

Materials can undergo reversible changes and irreversible changes

For example:

• some reversible changes to materials, such as dissolving, and changes of state are reversible as the product is not chemically altered

• most irreversible changes to materials, such as burning and rusting, produce new substances

• some irreversible changes to materials, such as cracking an egg or tearing paper, do not produce new substances

WA6SSUC1

Critical and creative thinking

Earth and space sciences

The effect of sudden geological events on Earth's surface, such as tsunamis, earthquakes and volcanic eruptions, and extreme weather, such as cyclones, extreme heat and floods

For example:

- the causes, magnitude and location of earthquakes, and where they most commonly occur, such as the Ring of Fire
- types of extreme weather, the factors that cause these and their effects, such as bushfires and floods

WA6SSUE1

Critical and creative thinking

Physical sciences

The transfer and transformation of energy in electrical circuits, including the role of circuit components, insulators and conductors

For example:

- the necessary components of an electrical circuit include a source, conductor and load, such as a globe or a buzzer
- closed circuits transfer electricity and transform electrical energy into different forms that include light, sound or motion
- common insulators and conductors and their practical applications in household appliances

WA6SSUP1

Critical and creative thinking

Science inquiry

Questioning and predicting

Pose testable questions that include variables to be measured and changed, and apply science knowledge to make predictions

For example:

- posing questions, such as
  - what happens to the brightness of globes in a circuit as more globes are added?
  - how is the growth of a bean sprout affected by the concentration of salt water it is immersed in?
- making predictions, such as
  - metals will conduct electricity, but plastics won't
  - more seeds will germinate in dark than in the light

WA6SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct fair, safe and repeatable investigations

For example:

- considering ways to approach investigations that include researching, trial-and-error, experimental testing, field observations, accessing digital tools to collect and manage data and virtual simulations
- making decisions on the variables to be controlled in fair tests

WA6SSIPL1

Literacy Critical and creative thinking Ethical understanding

Use equipment to observe, measure and record data

For example:

- using tools, such as digital thermometers or soil moisture probes, to collect and record data over time
- recording data in tables and diagrams or electronically, such as in spreadsheets

WA6SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Organise and represent data using tables, graphs and models to identify the relationships between measured and changed variables

For example:

- mapping the location of tectonic activity and representing the magnitude of earthquakes using a key
- graphing school electricity use to determine the times of the year when it is greatest
- representing the growth of plants in soils of varying salinity over time using appropriate graphs

WA6SSIPR1

Numeracy Critical and creative thinking

Evaluating

Compare findings with those of others, and to predictions; evaluate the fairness of an investigation and suggest improvements; and pose questions for further investigation

For example:

• discussing similarities and differences in their findings with those of others and/or their predictions

• recognising difficulties in conducting an investigation and planning for improvement

WA6SSIE1

Literacy Critical and creative thinking Personal and social capability

Communicating

Communicate ideas in a variety of ways, including scientific reports with appropriate language features

For example:

• preparing a scientific report to share findings about how plants respond to changes in physical conditions, such as temperature or salinity

• representing circuits using virtual simulations or circuit diagrams and indicating the direction of electricity flow

WA6SSICM1

Literacy Personal and social capability Digital literacy

Collaborating and applying

Use science knowledge to develop considered responses to problems, at a local and global level, through investigation and research

For example:

• researching structures resistant to seismic activity and the materials used in regions prone to such events to design a house that can withstand an earthquake

• collaborating with other students to research a local issue and suggest ways of solving it, such as ways to cool the classroom while reducing electricity usage

WA6SSICL1

Critical and creative thinking Personal and social capability

Year 7

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Teaching and learning programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Science provides opportunities for students to continue developing their understanding of important concepts and making connections between different areas of science and applications observed in their daily life.

In Year 7, students explore the diversity of life on Earth and continue to develop their understanding of the role of classification in ordering and organising information. They use and develop models, such as food chains and food webs, to represent energy flow in ecosystems and predict impacts of human activity. They use the particle theory to explain the motion and arrangement of atoms and molecules in the different states of matter and select appropriate techniques to separate pure substances from mixtures. They explore different types of celestial objects, investigate relationships in the Earth-Sun-Moon system and use models to predict and explain events. They consider the impact of forces acting on objects, represent and predict the effects of unbalanced forces on motion and determine the type of mechanical advantage provided by simple machines.

Students propose questions and make predictions based on scientific knowledge. They recognise risks when planning and conducting reproducible investigations. Students construct appropriate representations to organise and process data. They analyse data to describe patterns and relationships and use evidence to support conclusions. Students identify possible sources of error in their methods and suggest improvements. They use appropriate language and text features for their purpose and audience when communicating their ideas and findings. Students examine situations where development of scientific knowledge has benefited from collaboration and influenced the development of human activity.

Achievement Standard

By the end of the year:

Students use classification tools to classify and group organisms based on observable features. They represent the flow of energy in ecosystems and predict the impacts of human activity. They describe the motion and arrangement of atoms and molecules in solids, liquids and gases and describe techniques to separate pure substances in a mixture. They can classify celestial objects based on their observable properties and describe how the relative positions of Earth, the Sun and Moon affect phenomena on Earth. They can identify situations when friction, gravitational, magnetic and electrostatic forces are acting, represent and predict the effects of unbalanced forces on motion and identify the type of mechanical advantage provided by

simple machines.

Students plan and conduct reproducible investigations to test relationships and aspects of scientific models. They identify risks involved in conducting investigations. They use equipment to generate and record data with precision. They construct appropriate representations to organise data and information. They analyse data and information to describe patterns and relationships. They identify possible sources of error in methods and suggest improvements to their methods. They identify evidence to support their conclusions and support or dispute claims. They select and use language and text features appropriately for their purpose and audience when communicating their ideas and findings. They identify situations where development of scientific knowledge has benefited from collaboration and has influenced the development of human activity.

Content Descriptions

Science understanding

Biological sciences

Classification helps to order and organise the diversity of life on Earth into a hierarchy from kingdom to species; classification tools, including dichotomous keys, can be developed and used to classify organisms

For example:

• classifying organisms into taxonomic groups, such as organisms into kingdoms, arthropods and vertebrates into classes, mammals into monotremes, marsupials and placentals

• developing a dichotomous key to classify organisms in a local ecosystem

WA7SSUB1

Critical and creative thinking

Food chains and food webs can be used to represent energy flow in ecosystems and predict possible impacts of human activity

For example:

• role of photosynthesis and cellular respiration in the flow of energy in ecosystems

• feeding relationships in a local ecosystem

• predicting the possible impact of human activities that remove an organism from an ecosystem, such as overfishing, overuse of pesticides, habitat destruction and introduction of invasive species

• impact of introduced species on interactions in a local ecosystem, such as foxes, cats, cane toads and Phytophthora dieback

WA7SSUB2

Critical and creative thinking

Chemical sciences

Properties of the different states of matter can be explained by the motion and arrangement of atoms and molecules (particles); states can change with the addition or removal of energy

For example:

• comparing the motion and arrangement of atoms and molecules in solids, liquids and gases

• relating the properties of solids, liquids and gases, such as volume, shape, flow and compressibility to the motion and arrangement of atoms and molecules

• investigating the effect of heating and cooling on solids, liquids and gases

WA7SSUC1

Critical and creative thinking

Mixtures, including solutions, contain a combination of pure substances that can be separated based on their physical properties; a variety of techniques can be used to separate mixtures, including sieving, magnetic separation, decantation, filtration, evaporation, crystallisation, chromatography and distillation

For example:

• investigating saturated and unsaturated solutions and the solubility of different solutes

• selecting and using the appropriate techniques to separate

  • insoluble substances from a mixture, such as sand and pebbles from water

  • the solute and solvent from a solution, such as salt in water, pigments in felt pens

WA7SSUC2

Critical and creative thinking

Earth and space sciences

Celestial objects can be classified as planets, stars, moons, asteroids, meteoroids, comets, constellations and galaxies; planets in our solar system have distinguishing features, including composition, temperature, size, orbit, rotation, tilt of axis, moons and rings

For example:

• comparing the features of planets in our solar system

• researching Aboriginal and Torres Strait Islander peoples' knowledge of the night sky

WA7SSUE1

Literacy Critical and creative thinking

Predictable phenomena on Earth caused by its position relative to the Sun and the Moon, including lunar phases, eclipses, seasons and tides

For example:

• different regions on Earth experience different seasonal conditions

• tidal variations on Earth result from the relative positions of the Sun and Moon with respect to Earth

• exploring lunar eclipses and solar eclipses

• investigating Aboriginal and Torres Strait Islander peoples' calendars and how they predict seasonal changes

WA7SSUE2

Numeracy Critical and creative thinking

Physical sciences

Change to an object's motion is caused by unbalanced forces, including friction, gravitational, magnetic and electrostatic forces; the unit of measurement for force is the newton

For example:

• using arrows to represent the magnitude and direction of forces acting on an object

• friction can be increased by pressing surfaces together with more force and making the surface rougher

• friction can be decreased by reducing the pressure between surfaces, streamlining, polishing, lubricating and using rollers or ball bearings

• the weight of an object on Earth is the result of the gravitational attraction between its mass and that of the Earth

• exploring how electrostatic and magnetic forces cause motion

WA7SSUP1

Literacy Numeracy Critical and creative thinking

Simple machines, including levers, inclined planes and wheels and axles, provide a mechanical advantage, including force, distance and speed advantage

For example:

• the type of mechanical advantage provided by the different classes of levers, inclined planes and wheels and axles

• illustrating where simple machines are used every day

WA7SSUP2

Numeracy Critical and creative thinking

Science inquiry

Questioning and predicting

Propose investigable questions and make predictions based on scientific knowledge to explore scientific models, identify patterns and test relationships

For example:

• proposing investigable questions, such as

  • how does the angle of sunlight affect the surface temperature of Earth?

  • how does load carried affect the force of friction?

• making predictions, such as

  • if the closed seasons for fishing are removed, then the number of fish in the waterways will decrease

• if the temperature of a solvent is increased, then the solubility of a solute will increase

WA7SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct reproducible investigations to answer questions; recognise and manage risks and consider ethical issues

For example:

• planning and conducting a variety of investigation types, such as fair test investigations, descriptive investigations, comparative investigations and analytical investigations

• identifying risks and suggesting ways to manage them

• considering ethical issues, such as human consent, animal ethics, ecological ethics, use of heritage sites and artefacts on Country

WA7SSIPL1

Literacy Critical and creative thinking Ethical understanding

Select and use equipment to generate and record data with precision, using digital tools as appropriate

For example:

• selecting the appropriate equipment to collect data, such as using a measuring cylinder to measure volume

• recording data in appropriate formats, such as tables, spreadsheets and digital images, using appropriate units of measurement

• using digital tools, such as thermometers, electronic balances and force meters to collect data

WA7SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information

For example:

• constructing column graphs to represent discrete data and line graphs to represent continuous data

• processing data by calculating mean and percentages

• constructing models, such as solar system models, flow charts, force diagrams, dichotomous keys and simulations

WA7SSIPR1

Numeracy Critical and creative thinking

Analyse data and information to describe patterns and relationships, identify anomalies and draw conclusions based on evidence

For example:

• analysing data to describe patterns and relationships, such as the temperature on the surface of Earth is higher when the sun is directly overhead

• drawing logical conclusions considering the method of data collection, quality of evidence and limitations

WA7SSIPR2

Numeracy Critical and creative thinking

Evaluating

Reflect on scientific investigations, including evaluating the quality of the data collected, and identifying improvements

For example:

• reflecting on the quality of the data in relation to the data collection method and amount of data collected

• evaluating whether the investigation was valid and reliable

• suggesting improvements to investigation methods to increase the accuracy of the data recorded, such as using

  • the same electronic balance to measure the mass each time

  • a digital force meter to measure force rather than a spring balance

WA7SSIE1

Literacy Critical and creative thinking

Construct evidence-based arguments to support conclusions or evaluate claims

For example:

• using evidence provided by scientific investigations to evaluate claims or conclusions

• evaluating the quality of evidence when constructing an argument to support a conclusion or claim

• evaluating a claim that one brand of tyre is better on wet roads than another brand of tyre

WA7SSIE2

Numeracy Critical and creative thinking

Communicating

Communicate ideas, findings and information for specific purposes and audiences, including selection of appropriate content, language and text features, using digital tools as appropriate

For example:

• reporting on scientific investigations, incorporating diagrams, graphical representations and data as appropriate, and including examination of accuracy and reproducibility of the data

• writing a letter to the editor to express a view about the impact of spraying pesticides on a local ecosystem

• creating an informative text for a younger audience to show how the moon affects tides on Earth

WA7SSICM1

Literacy Critical and creative thinking Personal and social capability

Collaborating and applying

Illustrate how the development of scientific knowledge has benefited from collaboration across disciplines and the contributions of people from a range of cultures

For example:

• examining why it is important to recognise that different people in society have different perspectives on the introduction of biological controls to eradicate an invasive species

• investigating how aeronautical engineers' understanding of the nature of the forces acting in flight have led to changes in the design of aircraft

• exploring the contributions of Aboriginal and Torres Strait Islander peoples' knowledge to science, such as the identification of medicinal properties of endemic plants

• reflecting on the role of contemporary Aboriginal and Torres Strait Islander astronomers and astrophysicists, such as Wiradjuri astrophysicist and science communicator Kirsten Banks in promoting Aboriginal and Torres Strait Islander astronomy

WA7SSICL1

Critical and creative thinking

Illustrate how science understanding and skills have influenced the development of individual, community and workplace practices

For example:

• investigating everyday applications of physical separation techniques, such as desalination plants, sorting waste materials, reducing pollution, extracting products from plants, separating blood products and cleaning up oil spills

• exploring how tidal and seasonal changes affect people in a variety of activities, such as fishing and agriculture

• identifying endemic plants to revegetate a local ecosystem

• exploring how Dame Jane Goodall's communication of her research resulted in changed individual viewpoints and conservation policies

WA7SSICL2

Critical and creative thinking Personal and social capability

Year 8

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Teaching and learning programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Science provides opportunities for students to continue developing their understanding of important concepts and making connections between different areas of science and applications observed in their daily life.

In Year 8, students are introduced to cells and explore specialised flowering plant and vertebrate systems. They are also introduced to the structure of atoms and use representations to distinguish between elements and compounds. They classify elements as metals and non-metals based on their physical properties and distinguish between physical and chemical changes. They continue to develop a view of Earth as a dynamic system, exploring the interactions between processes occurring at plate boundaries and the rock cycle and how these processes explain patterns of change on Earth. They use physical properties to classify rocks and minerals and examine how properties of rocks reflect their formation and influence their use. They classify different forms of energy as kinetic or potential and represent energy transfer and transformation in simple systems. They explore in more detail how heat is transferred and electrical energy is transferred and transformed.

Students propose questions and make predictions based on scientific knowledge. They select appropriate equipment and manage risks when planning and conducting reproducible investigations. They select and construct appropriate representations to organise and process data. They analyse data to describe patterns and relationships and identify anomalies. They use evidence to support conclusions. They identify possible sources of error in their methods and suggest specific improvements. They select appropriate content, language and text features for their purpose and audience when communicating their ideas and findings. They examine situations where development of scientific knowledge has benefited from collaboration and influenced the development of human activity.

Achievement Standard

By the end of the year:

Students describe the role of cell structures. They describe the structure and function of flowering plant and vertebrate systems. They distinguish between elements and compounds and identify metals and non-metals based on their physical properties. They identify physical and chemical changes. They describe the geological features that form at plate boundaries. They describe the different processes of rock formation and classify rocks and minerals using their physical properties. They describe different forms of energy and represent transfer and transformation of energy in simple systems.

Students plan and conduct reproducible investigations to test relationships and aspects of scientific models. They describe and manage risks involved in conducting investigations. They select and use equipment to generate and record data with precision. They select and construct appropriate representations to organise data and information. They analyse data and information to describe patterns and relationships and identify anomalies. They identify possible sources of error in methods and suggest specific improvements to their methods. They construct evidence-based arguments to support conclusions and support or dispute claims. They select and use language and text features appropriately for their purpose when communicating their ideas, findings and information to specific audiences. They describe situations where development of scientific knowledge has

benefited from collaboration and has influenced the development of human activity.

Content Descriptions

Science understanding

Biological sciences

Cells are the basic units of living things and can be viewed with a compound microscope; cells contain organelles and structures with specialised functions; animal cells include the cell membrane, cytoplasm, nucleus and mitochondria; plant cells include the cell membrane, cytoplasm, nucleus, mitochondria, cell wall, chloroplasts and large vacuoles

For example:

• observing cell structures and determining cell size using a compound microscope

• photosynthesis occurs in the chloroplast of plant cells; (aerobic) cellular respiration occurs in the mitochondria of all plant and animal cells. Word equations can be used to represent these processes.

WA8SSUB1

Critical and creative thinking

Flowering plant and vertebrate systems carry out specialised functions that enable them to survive and reproduce, including systems for gas exchange, transportation of materials around the organism and reproduction

For example:

• comparing flowering plant and vertebrate systems for

  • gas exchange, such as stomata and guard cells in flowering plants and respiratory systems in vertebrates

  • transportation of materials, through the xylem and phloem, and processes of capillarity and transpiration in flowering plants and the heart, blood vessels and blood in vertebrates

  • reproduction, such as the structure of the flower and vertebrate reproductive systems, and the processes of pollination and fertilisation

WA8SSUB2

Critical and creative thinking

Chemical sciences

Matter is composed of atoms which contain protons, neutrons and electrons; matter can be classified as elements or compounds which can be compared using different representations, including symbols, formulae and models

For example:

• recalling the symbols for common elements

• using the formulae for molecules and compounds to identify the names and number/s of elements they contain

• constructing models of elements, molecules and compounds

WA8SSUC1

Critical and creative thinking

Elements of the periodic table can be classified as metals and non metals based on their physical properties

For example:

• exploring the properties of metals and non-metals, such as melting and boiling point, hardness, malleability, brittleness, ductility, heat and electrical conductivity, lustre and density

• comparing the properties of metals and non-metals, and their position on the periodic table

WA8SSUC2

Critical and creative thinking

Changes to substances can be classified as physical or chemical; chemical changes involve the formation of new substances

For example:

• identifying physical changes, such as changing state, dissolving, mixtures, tearing paper, chopping wood

• indicators of chemical changes, such as colour change, temperature change, formation of a precipitate, gas, odour, light

• identifying the reactants and products in a chemical reaction

WA8SSUC3

Critical and creative thinking

Earth and space sciences

The theory of plate tectonics explains global patterns of geological activity, including the formation of features at divergent, convergent and transform plate boundaries

For example:

• development of the theory of plate tectonics and evidence supporting the theory

• correlating earthquake and volcanic activity to tectonic plates

• examining why earthquakes occur within the Australian plate

WA8SSUE1

Critical and creative thinking

Rocks are composed of minerals; the key processes of the rock cycle are involved in the formation of igneous, sedimentary and metamorphic rocks; the properties of these rocks reflect their formation and influence their use

For example:

• classifying a range of common rocks using observable physical properties

• linking key processes of the rock cycle and the formation of rock types to processes within and between tectonic plates

• exploring the traditional geological knowledge of Aboriginal and Torres Strait Islander peoples in the selection of different rock types for different purposes

WA8SSUE2

Critical and creative thinking

Minerals can be classified using physical properties, including colour, streak, lustre, transparency, hardness and cleavage; useful resources can be extracted from minerals

For example:

• using a key to classifying minerals based on their physical properties

• exploring a local mineral resource, such as bauxite, diamonds

• exploring a local mineral resource

WA8SSUE3

Critical and creative thinking

Physical sciences

The different forms of energy can be classified as either kinetic or potential energy; energy transformations and transfers cause change within systems

For example:

• electrical, thermal, sound and light energy can be classified as kinetic energy

• gravitational, chemical, nuclear and elastic energy can be classified as potential energy

• energy transformations in systems, such as gravitational potential energy to kinetic energy, elastic potential energy in rubber bands or springs to kinetic energy, chemical energy to thermal energy

• energy transfers in systems, such as boiling water in a beaker, throwing a ball

WA8SSUP1

Critical and creative thinking

Heat is transferred by conduction in solids, convection in liquids and gases, and radiation in all states; heat can be reflected and absorbed

For example:

• comparing heat transfer via particle and wave models

• investigating conductors and insulators of heat

• exploring the absorption and reflection of heat by different surfaces, such as shiny and dull surfaces, dark- and light-coloured surfaces

WA8SSUP2

Critical and creative thinking

The flow of electricity through a circuit is affected by the type of circuit; a load placed in a circuit transforms electrical energy into other forms of energy; safety switches and circuit breakers are devices installed in buildings to protect people and electrical systems

For example:

• drawing circuit diagrams, using appropriate symbols, and constructing models of series and parallel circuits

• comparing safety switches and circuit breakers and researching regulations regarding their use

WA8SSUP3

Critical and creative thinking

Science inquiry

Questioning and predicting

Propose investigable questions and make predictions based on scientific knowledge to explore scientific models, identify patterns and test relationships

For example:

• proposing investigable questions, such as

  • how does car colour affect the internal temperature of a car?

  - how does temperature affect the amount of transpiration from plant leaves?
- making predictions, such as
  - if more globes are added to a circuit, then the light produced by each globe gets dimmer
  - if a saturated solution cools quickly, then the crystals formed will be smaller

WA8SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct reproducible investigations to answer questions; recognise and manage risks and consider ethical issues

For example:

- planning and conducting a variety of investigation types, such as fair test investigations, descriptive investigations, comparative investigations and analytical investigations
- identifying risks and suggesting ways to manage them
- considering ethical issues, such as human consent, animal ethics, ecological ethics, use of heritage sites and artefacts on Country

WA8SSIPL1

Literacy Critical and creative thinking Ethical understanding

Select and use equipment to generate and record data with precision, using digital tools as appropriate

For example:

- selecting the appropriate equipment to collect data, such as using a measuring cylinder to measure volume
- recording data in appropriate formats, such as tables, spreadsheets and digital images, using appropriate units of measurement
- using digital tools, such as thermometers, electronic balances, microscopes, voltmeters and ammeters to collect data

WA8SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information

For example:

- constructing column graphs to represent discrete data and line graphs to represent continuous data
- processing data by calculating mean, percentages and current
- constructing models, such as element, molecule and compound models, scale diagrams of cells, electrical circuit diagrams, the rock cycle, flow charts and simulations

WA8SSIPR1

Numeracy Critical and creative thinking

Analyse data and information to describe patterns and relationships, identify anomalies and draw conclusions based on evidence

For example:

- analysing data to describe patterns and relationships, such as darker coloured cars absorb more heat
- drawing logical conclusions considering the method of data collection, quality of evidence and limitations

WA8SSIPR2

Numeracy Critical and creative thinking

Evaluating

Reflect on scientific investigations, including evaluating the quality of the data collected, and identifying improvements

For example:

- reflecting on the quality of the data in relation to the data collection method and amount of data collected
- evaluating whether the investigation was valid and reliable
- suggesting improvements to investigation methods to increase the accuracy of the data recorded, such as using
  - the same electronic balance to measure the mass each time
  - a digital ammeter to measure current

WA8SSIE1

Literacy Critical and creative thinking

Construct evidence-based arguments to support conclusions or evaluate claims

For example:

- using evidence provided by scientific investigations to evaluate claims or conclusions
- evaluating the quality of evidence when constructing an argument to support a conclusion or claim

• evaluating a claim that one brand of battery lasts longer than another brand of battery

WA8SSIE2

Numeracy Critical and creative thinking

Communicating

Communicate ideas, findings and information for specific purposes and audiences, including selection of appropriate content, language and text features, using digital tools as appropriate

For example:

• reporting on scientific investigations, incorporating diagrams, graphical representations and data as appropriate, and including examination of accuracy and reproducibility of the data

• creating a documentary on the rock cycle, selecting appropriate language, models or analogies to engage a specific audience

• creating a digital infographic to illustrate heat loss through conduction, convection and radiation

WA8SSICM1

Literacy Critical and creative thinking Personal and social capability

Collaborating and applying

Illustrate how the development of scientific knowledge has benefited from collaboration across disciplines and the contributions of people from a range of cultures

For example:

• identifying how microscopes and medical imaging have led to improved understanding of cells and organs

• investigating how the development of super-strong, lighter alloys has enabled engineers to improve structural components in building, transportation and industry

• investigating how knowledge of the location and extraction of mineral resources relies on expertise from across disciplines

• exploring how seismic data is collected and shared between governments across the Asia-Pacific region and how governments use this data for alert systems, such as tsunami alerts

WA8SSICL1

Critical and creative thinking

Illustrate how science understanding and skills have influenced the development of individual, community and workplace practices

For example:

• exploring the development of spray-on skin by Professor Fiona Wood and Maria Stoner

• exploring importance of insect pollination in flowering plants for human food security

• examining the impact of mining on local ecosystems and the community

• exploring how geologist and oceanographic cartographer Marie Tharp's topographic maps of the Atlantic Ocean floor provided support for the acceptance of the theory of plate tectonics

WA8SSICL2

Critical and creative thinking Personal and social capability

Year 9

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Science provides opportunities for students to build on their understanding of important concepts and continue to develop scientific ideas and models to explain phenomena and events.

In Year 9, students explore how adaptations enable organisms to survive and respond to changes in their external environment and the impact of abiotic and biotic components on ecosystems. They use the structure and properties of atoms to determine the atomic and mass numbers for elements and the arrangement of elements on the periodic table. They use chemical formula to represent covalent and ionic compounds and chemical equations to represent chemical reactions. They explore how interactions within and between Earth's spheres affect the carbon cycle, water cycle and global climate. They begin to develop a more sophisticated view of energy transfer by exploring wave and particle models of energy transfer for light and sound.

Students propose questions and hypotheses to test relationships and develop models. They plan and conduct reproducible investigations and follow risk assessments when conducting investigations. They select and construct appropriate representations to organise, process and summarise data. They analyse data to describe patterns, relationships and anomalies and use a variety of evidence to support conclusions. They describe sources of error in methods and suggest ways to improve the quality of their data. They use content, language and text features to achieve their purpose when communicating their ideas, findings and arguments to specific audiences. They explore how advances in science, technologies and engineering are interconnected and examine how scientific responses impact society.

Achievement Standard

By the end of the year:

Students describe how plants and animals respond to external temperature changes and describe adaptations that enable them to survive. They describe the impact of abiotic and biotic factors on population size and species diversity. They describe the structure of atoms and the arrangement of elements in the periodic table. They write chemical formulae to represent simple compounds and word equations for chemical reactions. They describe how interactions within and between Earth's spheres affect the carbon cycle, water cycle and global climate. They describe wave and particle models of energy transfer for light and sound.

Students plan and conduct reproducible investigations to test or identify relationships and models. They follow risk assessments when conducting investigations. They select and use equipment to generate and record replicable data. They select and construct appropriate representations to organise, process and summarise data and information. They analyse data and information to identify patterns, relationships and anomalies. They describe sources of error in methods and suggest ways to improve the quality of their data. They draw conclusions that identify patterns or relationships evident in their data. They construct arguments based on evidence to support conclusions and evaluate claims. They use content, language and text features to achieve their purpose when communicating their ideas, findings and arguments. They identify how advances in science, technologies and engineering are interconnected and describe how scientific responses are developed and can impact society.

Content Descriptions

Science understanding

Biological sciences

Plants and animals have structural, behavioural and physiological adaptations that enable their survival in their environment

For example:

• plant adaptations to living in different ecosystems, such as the South West of Western Australia, deserts, tropical savannas and mangroves

• plant adaptations to bushfires

• vertebrate adaptations to living in different ecosystems, such as deserts, polar and marine ecosystems

WA9SSUB1

Critical and creative thinking

Organisms have mechanisms to respond to changes in their environment; endotherms and ectotherms respond differently to changes in external temperature; tropisms help plants respond to external stimuli

For example:

• how endotherms, such as mammals and birds, and ectotherms, such as reptiles and fish, use heat transfer mechanisms to respond to changes in external temperature

• plant tropisms, such as phototropism, geotropism and hydrotropism

WA9SSUB2

Critical and creative thinking

Population size and species diversity can be affected by abiotic and biotic factors; sampling techniques can be used to monitor abiotic factors and estimate numbers of organisms; ecological monitoring can be used to inform ecosystem health and impacts of human activity

For example:

• monitoring abiotic factors, such as temperature, pH, nutrients and salinity in a local ecosystem

• effect of biotic factors, such as competition, predation, symbiosis and human activity on population size and species diversity

• estimating population size in a local ecosystem using sampling techniques, such as capture/recapture and quadrats

• monitoring data to inform sustainable practices

WA9SSUB3

Numeracy Critical and creative thinking

Chemical sciences

The atomic number and mass number of an element can be used to determine the number of protons, neutrons and electrons in an atom of the element; isotopes of an element have the same number of protons but different numbers of neutrons in their nuclei and have the same chemical properties

For example:

• determining the number of protons, neutrons and electrons in atoms of different elements

• exploring the isotopes of elements, such as hydrogen and oxygen

WA9SSUC1

Numeracy Critical and creative thinking

The structure and properties of atoms relate to the organisation of the elements in the periodic table; elements in the same

group on the periodic table have similar properties

For example:

• exploring the periodic table to examine the similarities and differences between and within groups of elements, such as alkali metals, alkaline earth metals, transition metals, metalloids, non-metals, halogens and noble gases

• exploring the reactivity of metals with water through experimentation and relating reactivity to their position on the periodic table

• modelling the atomic structure of elements, including electron configuration

• modelling the structure of atoms from the same and different groups on the periodic table to determine structural similarities and differences

WA9SSUC2

Critical and creative thinking

Compounds are formed when electrons are transferred from one atom to another or are shared between atoms; non metal elements combine to form covalent substances; positively charged ions and negatively charged ions combine to form ionic compounds; compounds can be represented using chemical formulae and models

For example:

• recalling the chemical formulae for common covalent substances, such as water, carbon dioxide, oxygen gas and hydrochloric acid

• producing, collecting and testing covalent gases experimentally, such as oxygen, carbon dioxide and hydrogen

• using a table of common ions to write chemical formulae to represent ionic compounds, such as sodium chloride, magnesium chloride and iron (III) oxide

• exploring the conductivity of salt solutions experimentally

WA9SSUC3

Critical and creative thinking

Chemical reactions involve rearranging atoms to form new substances; word and balanced chemical equations can be used to represent the rearrangement of atoms in a chemical reaction and demonstrate the law of conservation of mass

For example:

• demonstrating the law of conservation of mass to model the balancing of chemical equations
  • experimentally
  • using molecular modelling kits, diagrams or simulations

• writing word equations and balanced chemical equations to demonstrate the law of conservation of mass when provided with the chemical formulae for reactants and products

• writing word equations and balanced chemical equations for chemical reactions observed in practical activities, such as the production of oxygen, carbon dioxide and hydrogen gas and reactivity of metals

WA9SSUC4

Critical and creative thinking

Earth and space sciences

Global systems, including the carbon and water cycles, rely on interactions involving the biosphere, lithosphere, hydrosphere and atmosphere

For example:

• the impact of human activity on the carbon cycle

• the impact of human activity on local freshwater availability

• main potable water resources for Western Australia, such as surface water, ground water and desalinated water

WA9SSUE1

Critical and creative thinking

Changes to global systems can be used to explain patterns of global climate change

For example:

• indicators of climate change, such as changes in oceanic and atmospheric temperatures, sea levels, species distribution, permafrost and sea ice coverage

• changes in global climate over time

WA9SSUE2

Numeracy Critical and creative thinking

Physical sciences

Sound waves are longitudinal waves produced by vibrating objects; sound waves travel through solids, liquids and gases at different speeds; sound is reflected when coming into contact with a solid or liquid surface

For example:

• exploring why we see lightning before we hear thunder even though they occur at the same time

• exploring resonance and its applications, such as musical instruments, shattering glass and bridge design

• exploring echoes and their applications, such as echolocation in nature, ultrasound in medicine, depth sounding in

oceanography and seismic surveying in mineral and oil exploration

• exploring the energy transfers that occur as sound travels through the ear and to the brain, and hearing conditions, such as conductive and nerve deafness

WA9SSUP1

Critical and creative thinking

Light is an electromagnetic wave; light is made up of photons that have both particle and wave properties; light can be reflected from plane and curved mirrors and refracted when passing through concave and convex lenses

For example:

• exploring the refraction of light as it passes from one medium to another

• investigating the size, nature and position of images formed by plane and curved mirrors, concave and convex lenses

• pathway of light entering the eye and focusing on the retina and corrections for eye defects, such as short-sightedness, long-sightedness and astigmatism

WA9SSUP2

Critical and creative thinking

Science inquiry

Questioning and predicting

Propose investigable questions and hypotheses to test relationships and develop explanatory models

For example:

• proposing investigable questions, such as
  • how do different lenses refract light?
  • how does gravity affect plant root and shoot growth?

• proposing hypotheses, such as
  • plants that live in hot, dry climates have fewer stomata in their leaves
  • sea ice reduces heat loss from oceans

WA9SSIQ1

Literacy Critical and creative thinking

Planning and conducting

Plan and conduct valid and reproducible investigations to answer questions and test hypotheses; develop and follow risk assessments, and consider ethical issues

For example:

• planning and conducting a variety of investigation types, such as fair test investigations, descriptive investigations, comparative investigations and analytical investigations

• outlining strategies to control possible sources of systematic errors and random errors

• developing risk assessments to identify potential hazards and prevent potential incidents and injuries

• considering ethical issues, such as human consent, animal ethics, ecological ethics, use of heritage sites and artefacts on Country

WA9SSIPL1

Literacy Critical and creative thinking Ethical understanding

Select and use equipment to generate and record data with precision to obtain appropriate sample sizes and replicable data, using digital tools as appropriate

For example:

• recording data in appropriate formats, such as tables, spreadsheets and digital images, using appropriate units of measurement

• obtaining reliable data through a large sample size, replicates and repeating an experiment

• using digital tools, such as thermometers, electronic balances, pH meters, light meters and sound meters to collect data

WA9SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Select and construct appropriate representations, including tables, graphs, descriptive statistics, models and mathematical relationships, to organise and process data and information

For example:

• constructing column graphs to represent discrete data and line graphs to represent continuous data

• processing data by calculating percentages and ratios, and using descriptive statistics, such as mean, mode, median and range

• constructing models, such as chemical formulae, word and chemical equations, biogeochemical cycles, light ray diagrams,

flow charts and simulations

WA9SSIPR1

Numeracy Critical and creative thinking

Analyse and connect a variety of data and information to identify and explain patterns, relationships and anomalies, and draw conclusions based on evidence

For example:

• analysing data to describe patterns and relationships, such as plant shoots grow upwards and roots grow downwards, and explaining patterns and relationships using relevant science concepts

• writing conclusions that describe patterns or relationships and providing evidence from the investigation and research to support the conclusion

WA9SSIPR2

Numeracy Critical and creative thinking

Evaluating

Evaluate validity and reliability of methods and validity of conclusions, including identifying possible sources of error, and describe specific ways to improve the quality of the data

For example:

• comparing the data collected with the expected values and comment on the accuracy of the data
• reflecting on the precision of the data collected and identifying sources of error
• linking the quality of the data to the validity of the conclusion
• suggesting specific improvements to the validity of the investigation, such as
  • controlling all specific variables
  • using a control for comparison
  • increasing random samples
• suggesting specific improvements to the reliability of the investigation, such as
  • ensuring consistency of results
  • having a large enough sample
  • repeating or replicating the investigation

WA9SSIE1

Literacy Critical and creative thinking

Construct arguments based on analysis of a variety of evidence to support conclusions or evaluate claims

For example:

• constructing an argument supported by primary and secondary evidence and reasoning to support or reject hypotheses
• evaluating the quality of primary and secondary evidence when constructing an argument to support a conclusion or claim
• constructing a scientific argument showing how a range of evidence supports a claim relating to climate change

WA9SSIE2

Numeracy Critical and creative thinking

Communicating

Communicate scientific ideas and information for specific purposes and audiences, including constructing evidence-based arguments and selection of appropriate content, language and text features, using digital tools as appropriate

For example:

• reporting on investigations, incorporating background information, diagrams, graphical representations and data as appropriate, an explanation of the results using scientific knowledge, and a discussion that considers validity and reproducibility

• summarising the findings of an investigation as a live or virtual poster presentation

• creating a digital infographic to highlight the multiple lines of evidence from polar ice caps, ocean temperatures and extreme weather to explain how climate change is impacting Earth

WA9SSICM1

Literacy Critical and creative thinking Personal and social capability

Collaborating and applying

Illustrate how advances in scientific understanding often rely on developments in technologies and engineering and technological and engineering advances are often linked to scientific discoveries

For example:

• investigating how computer modelling can be used to predict the changes in populations due to environmental changes, such as the impact of flooding or fire on rabbit or kangaroo populations

• considering how computer modelling has improved knowledge and predictability of phenomena, such as climate change and atmospheric pollution

• exploring Australian technological advances, such as the cochlear implant pioneered by Professor Graeme Clark and the Monash Vision Group's work on a bionic eye

WA9SSICL1

Critical and creative thinking

Illustrate how proposed scientific responses to contemporary issues may impact on society

For example:

• investigating strategies implemented to maintain part of the local environment, such as bushland, a beach, a lake, a desert or a shoreline

• exploring how Marie Curie's discovery of new elements and work with radioisotopes contributed to the treatment of cancers

• examining climate change mitigating technologies to reduce energy consumption, such as electric vehicles, solar panels and battery storage by individuals, industries and communities

• considering the availability and impacts, such as environmental, ethical, social and economic considerations, of using different renewable and non renewable energy resources to produce electricity

• considering safe sound levels for humans and implications in the workplace and leisure activities

WA9SSICL2

Critical and creative thinking Personal and social capability

Year 10

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Science provides opportunities for students to build on their understanding of important concepts and continue to develop scientific ideas and models to explain phenomena and events.

In Year 10, students investigate processes that underpin heredity and natural selection to understand the continuity of life. They develop a more sophisticated understanding of atomic theory and explore patterns and relationships within the periodic table to explain ionic and covalent bonding. They predict the effect of changing reactant and reaction conditions, use patterns of reactions to predict reaction products and represent reactions using chemical equations. They explore the key events in the formation of stars, galaxies and planetary systems and how space exploration has contributed to knowledge of the formation and evolution of the universe and improved life on Earth. They understand that motion and forces are related by applying physical laws and mathematical models. They apply the law of conservation of energy to analyse system efficiency.

Students propose questions and hypotheses to test relationships and develop models. They develop and follow risk assessments when planning and conducting reproducible investigations with appropriate sample sizes and replicable data. They select and construct appropriate representations to organise, process and summarise data. They analyse data to describe patterns, relationships and anomalies and use a variety of evidence to support conclusions. They consider the validity and reliability of methods and suggest ways to improve the quality of the data. They use content, language and text features to achieve their purpose when communicating their ideas, findings and arguments to specific audiences. They explore how advances in science, technologies and engineering are interconnected and examine how scientific responses impact society.

Achievement Standard

By the end of the year:

Students describe the process of inheritance in terms of genes and predict the outcome of autosomal dominant/recessive crosses. They describe the processes involved in the theory of evolution by natural selection. They describe how the periodic table organises elements and use it to make predictions about types of bonding. They predict the effect of changing reactant and reaction conditions and use patterns of reactions to predict reaction products. They sequence key events in the formation of stars, galaxies and planetary systems and describe the benefits of space exploration. They recall Newton's laws of motion and use them to predict motion of objects in a system. They describe the concept of energy conservation and represent energy transfer and transformation within systems.

Students plan and conduct valid and reproducible investigations to test or identify relationships and models. They follow risk assessments when conducting investigations. They select equipment and use it to generate and record appropriate sample sizes and replicable data. They select and construct appropriate representations to organise, process and summarise data and information. They analyse data and information to identify and describe patterns, relationships and anomalies. They describe the validity and reliability of methods and suggest ways to improve the quality of the data. They use evidence to draw conclusions that identify patterns or relationships evident in their data. They construct arguments based on analysis of a variety of evidence to support conclusions and evaluate claims. They use content, language and text features effectively to achieve their purpose when communicating their ideas, findings and arguments to specific audiences. They describe how advances in science, technologies and engineering are interconnected and analyse scientific responses and how these responses impact society.

Content Descriptions

Science understanding

Biological sciences

Cell division processes of meiosis and mitosis produce new cells with chromosome numbers specific to their role; chromosomes contain genes that are composed of DNA (deoxyribonucleic acid)

For example:

- comparing the role and number of chromosomes in the cells produced by mitosis and meiosis
- modelling the structure of DNA

WA10SSUB1

Critical and creative thinking

Patterns of monohybrid inheritance, including autosomal dominant/recessive and sex-linked recessive inheritance, can be predicted using pedigrees and Punnett square crosses

For example:

- interpreting pedigrees and predicting offspring for
  - autosomal dominant/recessive alleles, such as hair colour in guinea pigs, leaf colour in barley, seed shape and colour, and plant height in peas
  - sex-linked recessive alleles, such as red--green colour blindness and haemophilia

WA10SSUB2

Numeracy Critical and creative thinking

The theory of evolution by natural selection explains the past and present diversity of living things, including variation within a species, adaptations and speciation

For example:

- natural selection can be used to explain
  - differences in organisms of the same species at different locations, such as colour, size and shape
  - adaptations of organisms to changing environments, such as colour, size, and resistance to antibiotics and pesticides
  - how isolation can result in one species becoming a different species over time, such as Australian marsupials or Galápagos finches
- the importance of maintaining genetic diversity

WA10SSUB3

Critical and creative thinking

Chemical sciences

The ability of atoms to form chemical bonds can be explained by the arrangement of electrons in the atom; ionic bonding involves electron transfer and covalent bonding involves sharing of electrons

For example:

- comparing ionic and covalent bonds
- using the naming conventions for ionic and covalent compounds
- representing compounds using chemical formulae and models
- using the periodic table and a table of common ions to write chemical formulae to represent ionic compounds

WA10SSUC1

Critical and creative thinking

Reactions follow general patterns that help to predict the reaction products, including precipitation reactions and reactions of acids with bases, metals and carbonates; word and balanced chemical equations can be used to represent these reactions

For example:

- modelling the balancing of chemical equations using molecular modelling kits, diagrams or simulations
- using indicators to identify acids and bases, and observing the changes in neutralisation reactions
- everyday applications of
  - precipitation reactions, such as water testing
  - acid reactions, such as indigestion, fire extinguishers and the treatment of acidic soil and bee stings

WA10SSUC2

Critical and creative thinking

The rate at which a reaction occurs can be altered by changing factors, including temperature, concentration and the surface area of a reactant

For example:

- investigating factors affecting the rate of reactions, such as changing the
  - temperature of the reactants
  - concentration of one or more reactants
  - surface area of a solid reactant

WA10SSUC3

Critical and creative thinking

Earth and space sciences

The formation of stars, galaxies and planetary systems has continued since the time of the Big Bang; stars have a life cycle determined by their mass

For example:

• the life cycle of different types of stars
• the formation of the Milky Way galaxy
• the nebular hypothesis for the formation of our solar system

WA10SSUE1

Critical and creative thinking

Space exploration contributes to knowledge of the formation and evolution of the universe and Earth, as well as providing useful tools and technologies to improve our life on Earth

For example:

• data gathered through space rover and probe missions is providing information about the geology and atmosphere of moons and other planets in our solar system
• data gathered through space exploration can be used to search for possible life beyond our solar system
• applications of space technology, including remote sensing of Earth, communication systems and scientific exploration
• importance of Australia's role in space exploration
• contribution of the Square Kilometre Array (SKA) to space exploration
• impact of space junk and satellites on future space exploration and Earth
• impact of low gravity on biological systems

WA10SSUE2

Critical and creative thinking

Physical sciences

Motion can be quantitatively determined; quantities, including time, distance, displacement, speed, velocity and acceleration, can be classified as scalar or vector; vector diagrams can be used to represent the magnitude and direction of motion

For example:

• using vector diagrams to illustrate the difference between distance and displacement
• exploring the relationship between time intervals and distance or displacement to calculate speed, velocity and acceleration

WA10SSUP1

Numeracy Critical and creative thinking

Newton’s laws of motion can be used to predict motion; the relationship between force, mass and acceleration of objects can be quantitatively determined

For example:

• investigating Newton's three laws of motion
• applying Newton's three laws of motion to everyday applications, such as sport and driving

WA10SSUP2

Numeracy Critical and creative thinking

The law of conservation of energy can be applied to analyse system efficiency in terms of energy inputs and outputs, transfers and transformations

For example:

• work is done when energy is transferred and transformed
• energy and work are scalar quantities that can be quantitatively determined by calculating gravitational potential and kinetic energy
• modelling the efficiency of energy transfers and transformations in a system using Sankey diagrams
• calculating energy transformations in a system, such as a ball falling onto a surface and rebounding, a pendulum swing, a roller coaster changing height and speed around its track

WA10SSUP3

Numeracy Critical and creative thinking

Science inquiry

Questioning and predicting

Propose investigable questions and hypotheses to test relationships and develop explanatory models

For example:

• proposing investigable questions, such as
  • how do safety devices, such as seatbelts, reduce the likelihood of injury in a collision?
  • are traits caused by dominant or recessive alleles more common in the class?

- proposing hypotheses, such as
  - the rate of a reaction increases as the temperature of reactants increases
  - the steeper the slope of a ramp the greater the acceleration of an object down the slope

WA10SSIQ1

Numeracy Critical and creative thinking

Planning and conducting

Plan and conduct valid and reproducible investigations to answer questions and test hypotheses; develop and follow risk assessments, and consider ethical issues

For example:

- planning and conducting a variety of investigation types, such as fair test investigations, descriptive investigations, comparative investigations and analytical investigations
- outlining strategies to control possible sources of systematic errors and random errors
- developing risk assessments to identify potential hazards and prevent potential incidents and injuries
- considering ethical issues, such as human consent, animal ethics, ecological ethics, use of heritage sites and artefacts on Country

WA10SSIPL1

Literacy Critical and creative thinking Ethical understanding

Select and use equipment to generate and record data with precision to obtain appropriate sample sizes and replicable data, using digital tools as appropriate

For example:

- recording data in appropriate formats, such as tables, spreadsheets and digital images, using appropriate units of measurement
- obtaining reliable data through a large sample size, replicates and repeating an experiment
- using digital tools, such as thermometers, electronic balances, pH meters, microscopes and motion sensors to collect data

WA10SSIPL2

Numeracy Digital literacy

Processing, modelling and analysing

Select and construct appropriate representations, including tables, graphs, descriptive statistics, models and mathematical relationships, to organise and process data and information

For example:

- constructing column graphs to represent discrete data and line graphs to represent continuous data
- processing data by calculating percentages, ratios, speed, velocity, acceleration and force, and using descriptive statistics, such as mean, mode, median and range
- constructing models, such as atomic models, chemical formulae, word and chemical equations, pedigrees, vector diagrams, flow charts and simulations

WA10SSIPR1

Numeracy Critical and creative thinking

Analyse and connect a variety of data and information to identify and explain patterns, relationships and anomalies, and draw conclusions based on evidence

For example:

- analysing data to describe patterns and relationships, such as the steeper the slope of a ramp the greater the acceleration of an object as it travels down the slope, and explaining patterns and relationships using relevant science concepts
- writing conclusions that describe patterns or relationships and providing evidence from the investigation and research to support the conclusion

WA10SSIPR2

Numeracy Critical and creative thinking

Evaluating

Evaluate validity and reliability of methods and validity of conclusions, including identifying possible sources of error, and describe specific ways to improve the quality of the data

For example:

- comparing the data collected with the expected values and comment on the accuracy of the data
- reflecting on the precision of the data collected and identifying sources of error
- linking the quality of the data to the validity of the conclusion
- suggesting specific improvements to the validity of the investigation, such as
  - controlling all specific variables
  - using a control for comparison
  - increasing random samples

- suggesting specific improvements to the reliability of the investigation, such as
  - ensuring consistency of results
  - having a large enough sample
  - repeating or replicating the investigation

WA10SSIE1

Literacy Critical and creative thinking

Construct arguments based on analysis of a variety of evidence to support conclusions or evaluate claims

For example:

- constructing an argument supported by primary and secondary evidence and reasoning to support or reject hypotheses
- evaluating the quality of primary and secondary evidence when constructing an argument to support a conclusion or claim
- constructing a scientific argument showing how a range of evidence supports the theory of natural selection

WA10SSIE2

Numeracy Critical and creative thinking

Communicating

Communicate scientific ideas and information for specific purposes and audiences, including constructing evidence-based arguments and selection of appropriate content, language and text features, using digital tools as appropriate

For example:

- reporting on scientific investigations, incorporating background information, diagrams, graphical representations and data as appropriate, an explanation of the results using scientific knowledge, and a discussion that considers validity and reproducibility
- summarising the findings of an investigation as a live or virtual poster presentation
- using animation or comic strip software to create an explanation of the life cycle of a star for an audience of their peers

WA10SSICM1

Literacy Critical and creative thinking Personal and social capability

Collaborating and applying

Illustrate how advances in scientific understanding often rely on developments in technologies and engineering and technological and engineering advances are often linked to scientific discoveries

For example:

- exploring how the development of fast computers has made the analysis of DNA sequencing possible
- examining how the work of Rosalind Franklin, James Watson, Francis Crick and Maurice Wilkins contributed to the development of the double helix structure of DNA
- exploring how astronomer Vera Rubin's discovery of the existence of dark matter was validated
- exploring how the development of fast computers has made possible the analysis of radio astronomy signals and other data generated by major international science projects, such as the Event Horizon Telescope and the Square Kilometre Array (SKA)
- examining how the discovery of gravity waves validated Einstein's theory of general relativity and why this discovery did not occur until 100 years after the theory was proposed

WA10SSICL1

Critical and creative thinking

Illustrate how proposed scientific responses to contemporary issues may impact on society

For example:

- using knowledge of science to test claims made in advertising
- examining karyotypes and applications of biotechnologies, such as DNA profiling, gene therapy and genetic engineering
- investigating why agricultural practices have changed to include widespread use of genetically engineered crops
- examining how the recent use of female crash test dummies has shown women are at greater risk of injury in a car accident and consider implications for changing car safety features
- describing how science is used in the media to explain a natural event or justify people's actions

WA10SSICL2

Critical and creative thinking Personal and social capability